

vaadin}>

Full-Stack Productivity: Web Apps in Java with Vaadin



Jürgen Tremel
Customer Success



2 DAYS AGO ON HN

React is winning by default and slowing innovation

2 DAYS AGO ON HN

When teams need a new frontend,
the conversation rarely starts with
"What are the constraints and which tool best fits them?"
It often starts with
"Let's use React; everyone knows React."

That reflex creates a self-perpetuating cycle where network effects, rather than technical fit, decide architecture.

BRIEF & DISCLAIMER

Who's talking?

20+ years in web development

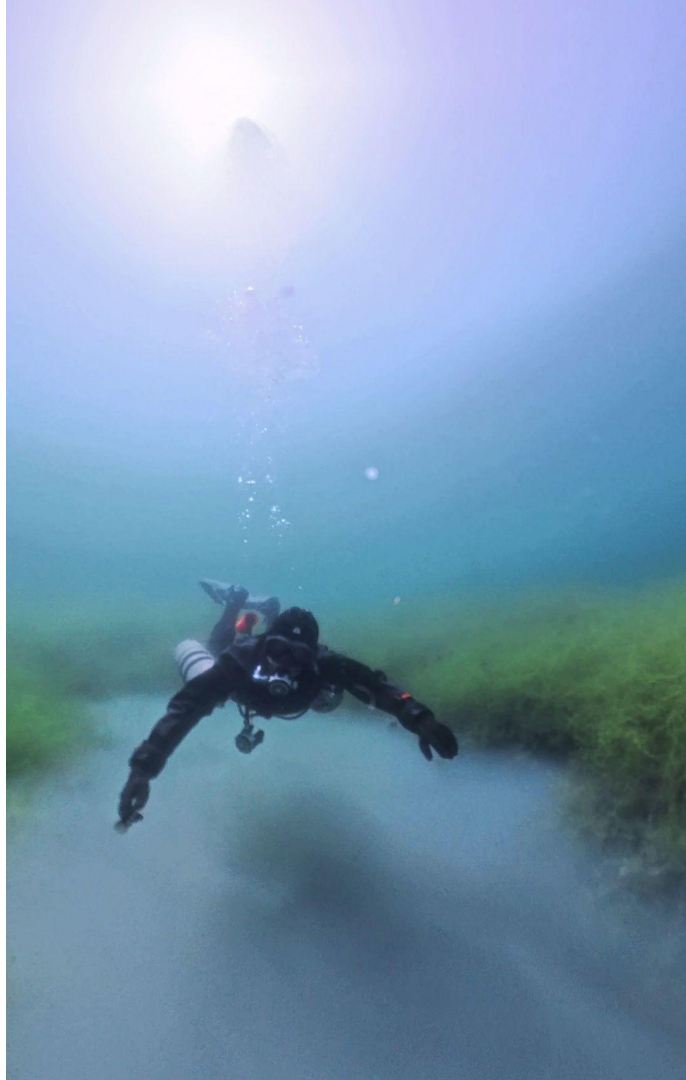
Originally aerospace R&D

10+ years at Vaadin

Cross-industry

Currently Customer Success
and Solution Architecture

Off-work mostly underwater



Vaadin

Platform & Company

OPEN-SOURCE & FREE

Core Java Framework

High-quality set of UI components

Collaboration Kit

Figma templates

3rd-party component directory

COMMERCIAL TOOLS & SERVICES

Additional components (Charts, Maps, Dashboard, ...)

Tools (Copilot, Testbench, ...)

Kits for SSO, Observability, Kubernetes, ...

Swing & Legacy Modernization (Automation Tooling, Components)

Support & Development Services

Commercial Maintenance

01

Vaadin Birds-Eye View

Who's using it?

How is it different?

How does it work?

02

Live Demo

Vaa... what?

Northern Origins & Bridging the Gap

vaadin>

Finnish for (female) reindeer

The birthplace of the Vaadin framework

Bringing `<web applications>`
to the `Java {}` world.



Who's using it?

We've built business web applications for many business verticals since the year 2000.

81%

Of Fortune
500 Companies

25+

Years of
Experience

200k+

Developers
worldwide

The hard life of a web developer

FAST-MOVING

Framework “of the day”

Myriad toolchains

Both evolve and become outdated fast

SKILL SETS & SILOS

Coordinate designers, frontend and backend teams

Agree APIs between frontend and backend

All requiring different skill sets

Steep learning curves and on-boarding

CUTTING-EDGE & EXCITING - MAYBE.

PRAGMATIC, FAST, EFFICIENT - NOT SO MUCH!

How things could be

ONE STACK, FULL STACK

AUTOMATED COMMUNICATION

SOLID FOUNDATION & ECOSYSTEM

REUSE OF TALENT AND SKILLS

FAST & EFFICIENT BY DEFAULT

FLEXIBLE & CUSTOMIZABLE WHEN NEEDED

Building web apps in pure Java

Full-stack with Vaadin

 Comes with **50+ ready-to-use UI components**

Labels, Button, Grids,
Charts, Maps, ...

 Fully integrated with
the **Java Ecosystem**

Works with Spring Boot,
Maven, JUnit, ...

 **Java only**, no need
for HTML, CSS or
JavaScript

But customizable when
needed

 Using **modern web technologies** under
the hood

NPM, Vite &
WebComponents

 **Enhanced security**
out-of-the-box

Server-side logic, state,
data binding and validation
reduce attack surfaces.

 **Easy deployment**
in different
environments

Ship as jar or war, just
like any Java app

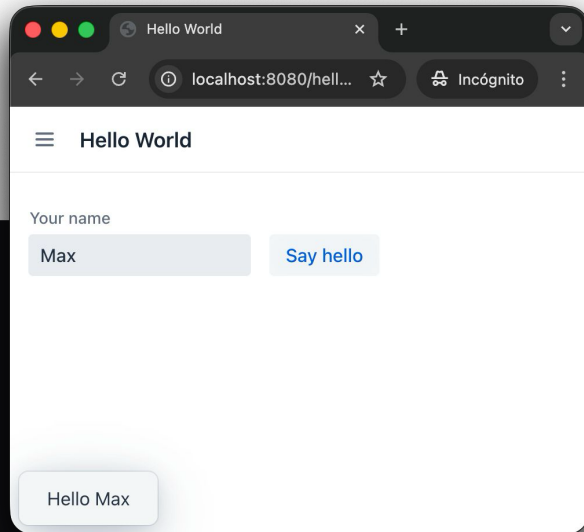
Building web apps in pure Java

Vaadin *Hello World*

```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView() {
        var name = new TextField("Your name");
        var sayHello = new Button("Say hello", event ->
            Notification.show("Hello " + name.getValue()));

        add(name, sayHello);
        setMargin(true);
    }
}
```



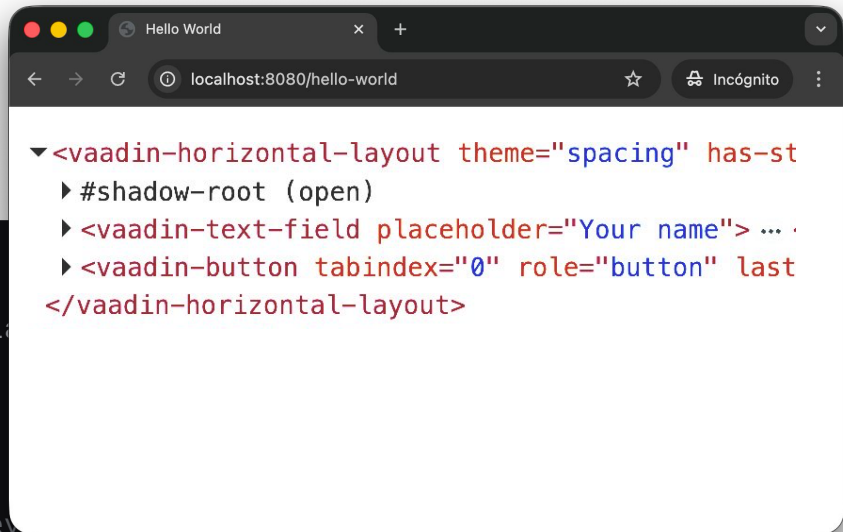
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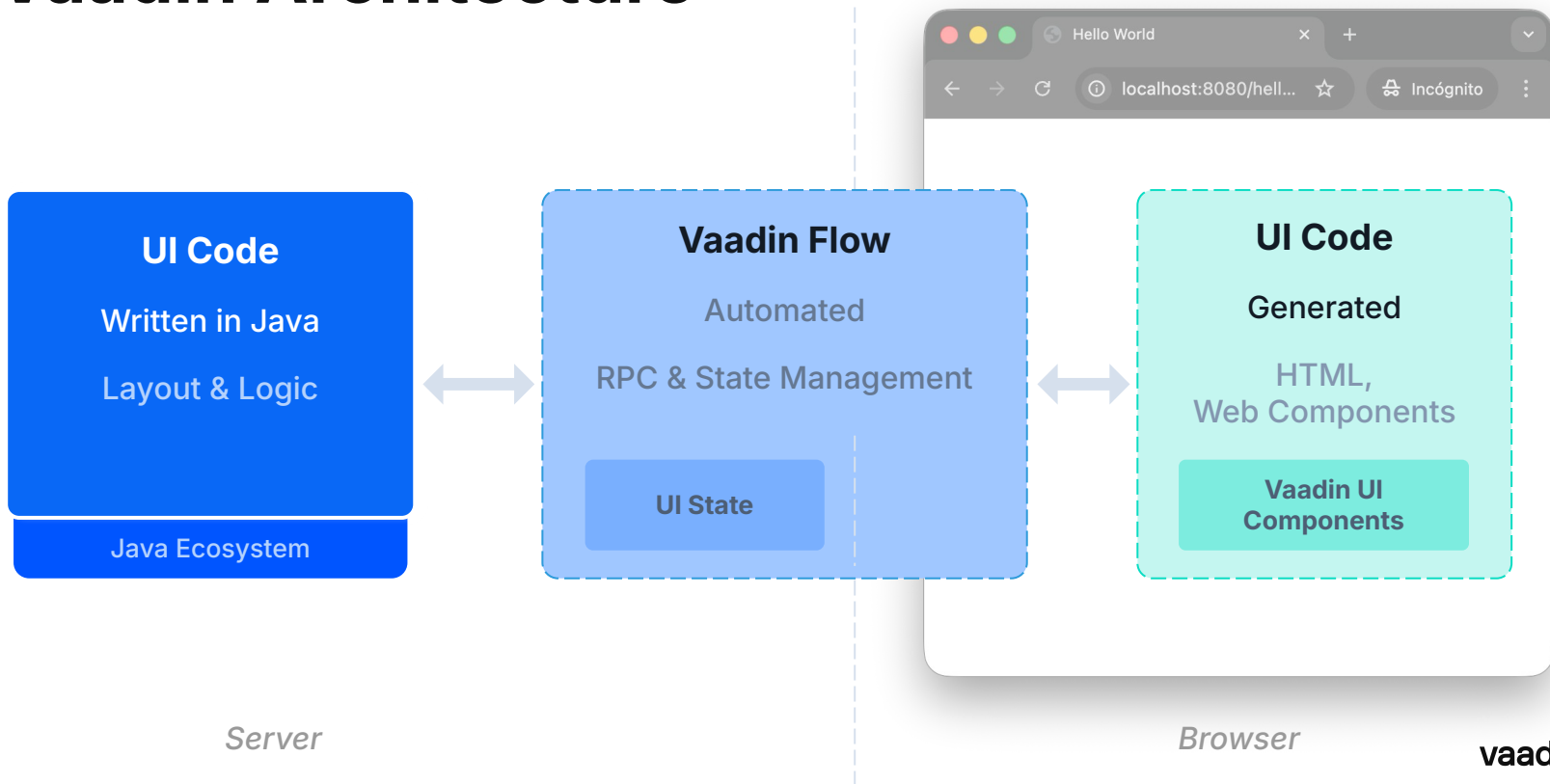
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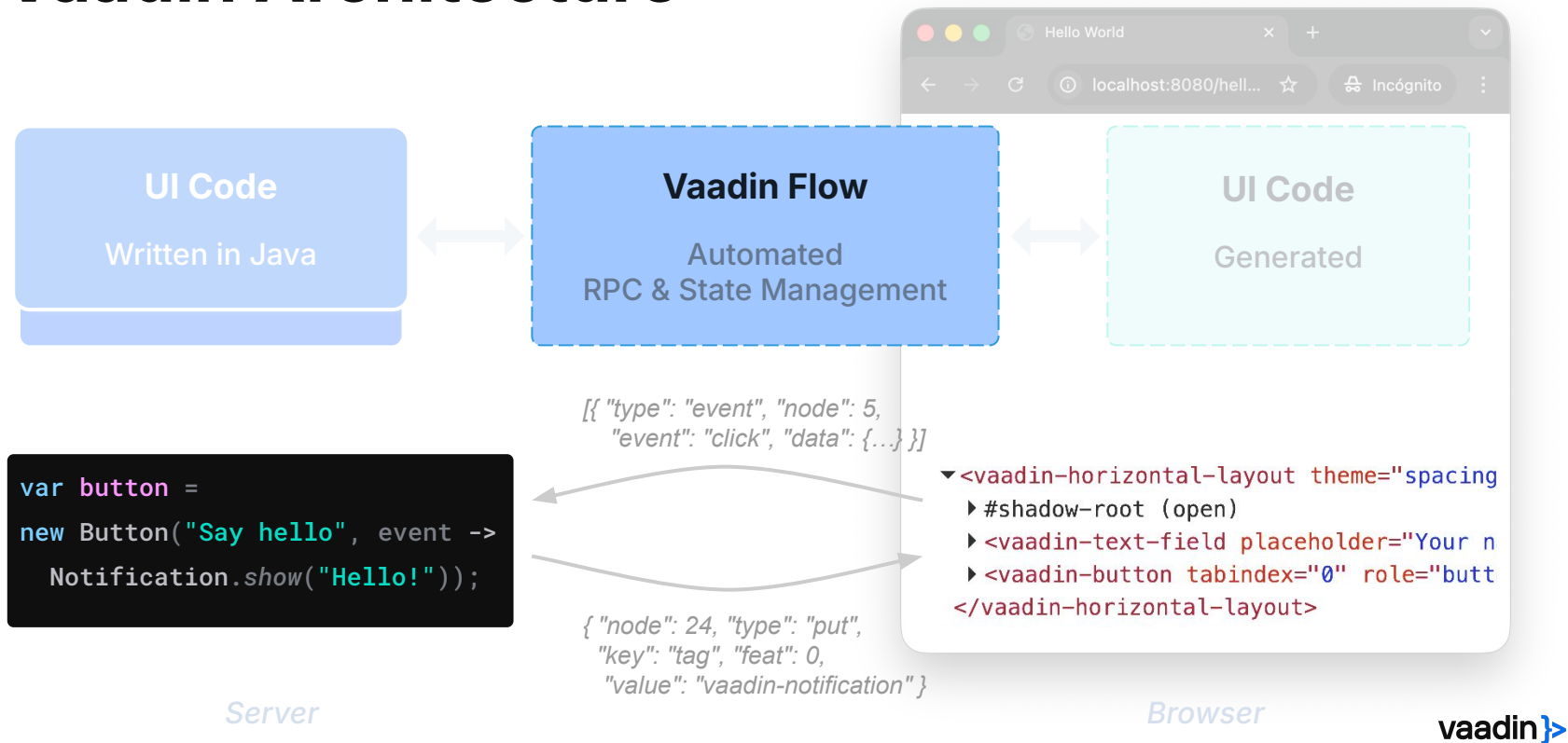


Building web apps in pure Java

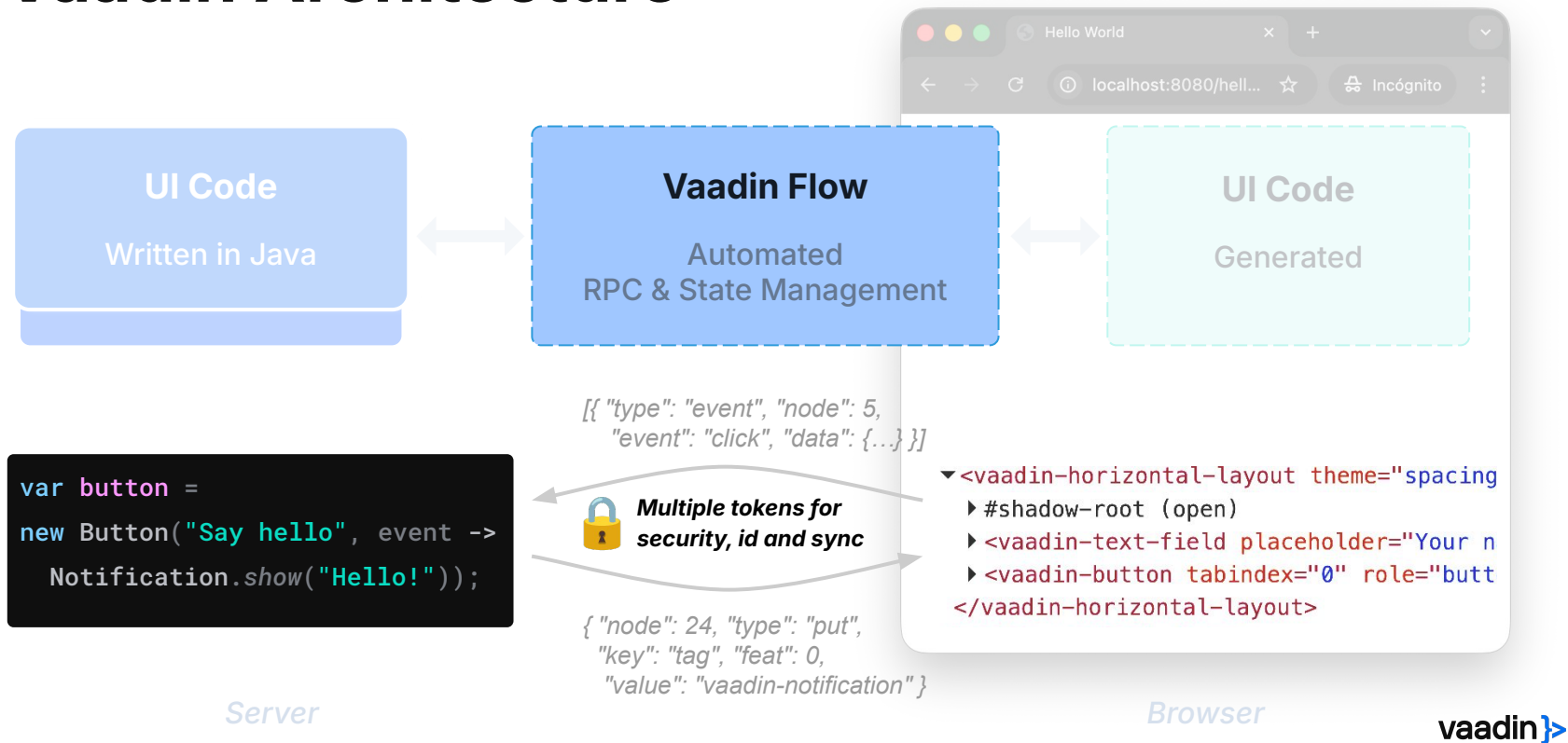
Vaadin Architecture



Vaadin Architecture



Vaadin Architecture



01

Vaadin Birds-Eye View

02

Live Demo

Hello World

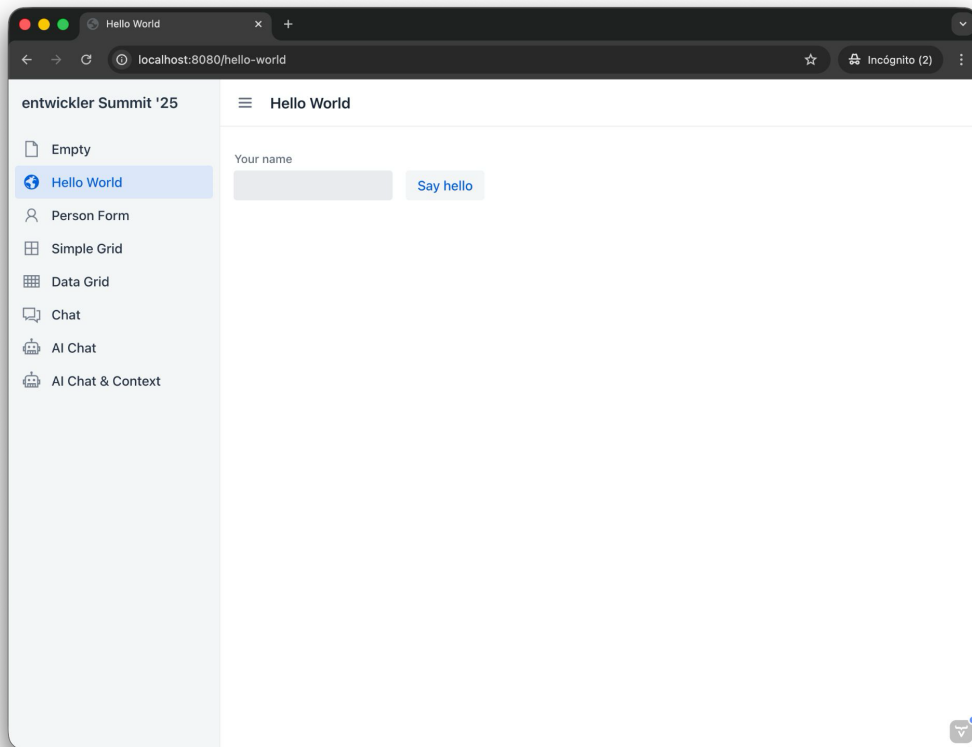
Theming & Visual Editing

Forms & Data Grids

Chat, Collaboration & AI

DEMO I

Hello World



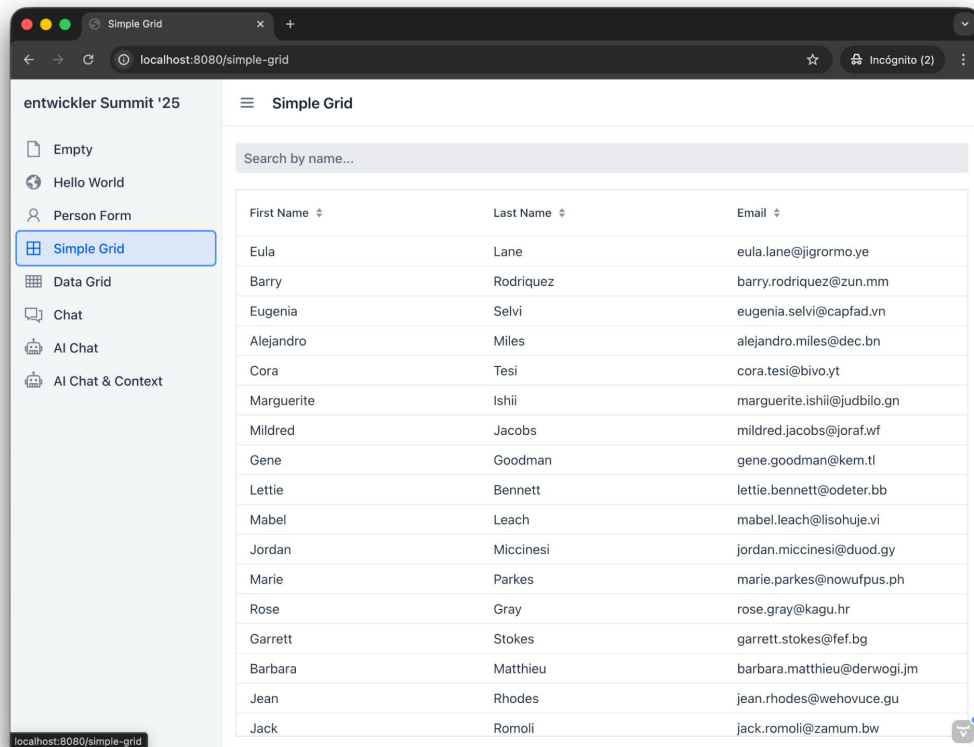
DEMO II

Forms & Form Layout

The screenshot shows a web browser window with the address bar displaying 'localhost:8080/person-form'. The browser's address bar also shows 'Incognito (2)'. The application has a sidebar on the left with the title 'entwickler Summit '25'. The sidebar contains a list of items: 'Empty', 'Hello World', 'Person Form' (highlighted with a blue border), 'Simple Grid', 'Data Grid', 'Chat', 'AI Chat', and 'AI Chat & Context'. The main content area is titled 'Person Form' and contains a section titled 'Personal Information'. This section has six input fields: 'First Name', 'Last Name', 'Birthday' (with a calendar icon), 'Phone Number', 'Email', and 'Occupation'. At the bottom of the form are two buttons: 'Save' (blue) and 'Cancel' (gray). The bottom right corner of the application window has a small icon.

DEMO III

Simple Grid & Search



Advanced Grid

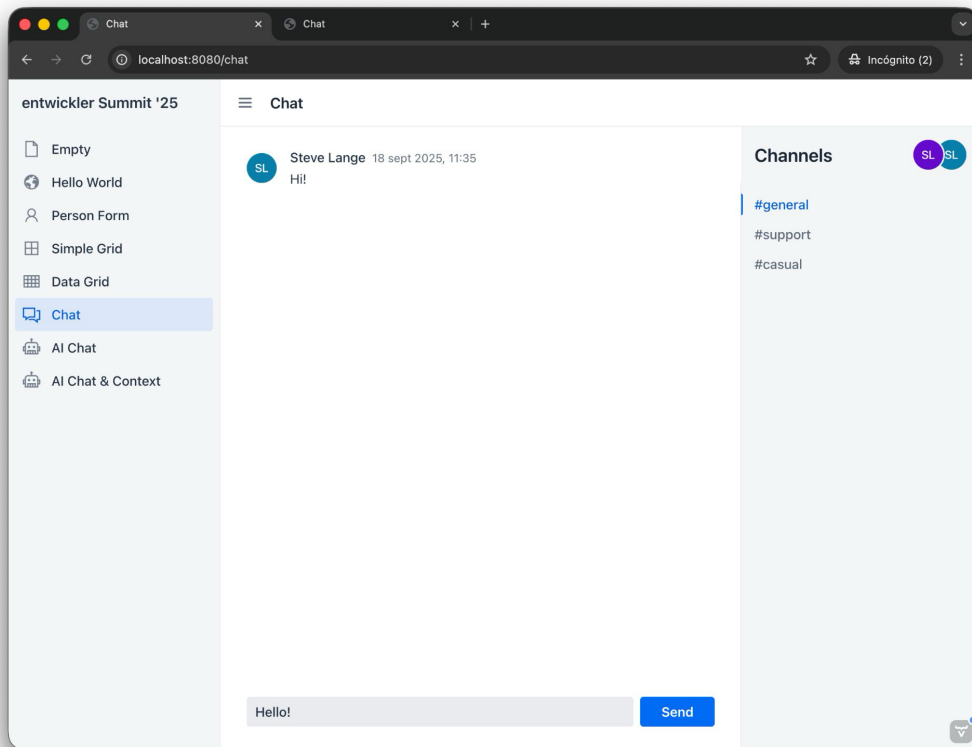
entwickler Summit '25

Data Grid

Client	Amount	Status	Date
Filter	Filter	Filter	Filter
Amarachi Nkechi	\$47,427.00	Success	5/9/2019
Bonelwa Ngqawa...	\$70,503.00	Success	5/9/2019
Debashis Bhuiyan	\$58,931.00	Success	5/7/2019
Jacqueline Asong	\$25,053.00	Pending	4/25/2019
Kobus van de Ve...	\$7,319.00	Pending	4/22/2019
Mattie Blooman	\$18,441.00	Error	4/17/2019
Oea Romana	\$33,376.00	Pending	4/17/2019
Stephanus Huggi...	\$75,774.00	Success	2/26/2019
Torsten Paulsson	\$82,531.00	Pending	2/21/2019

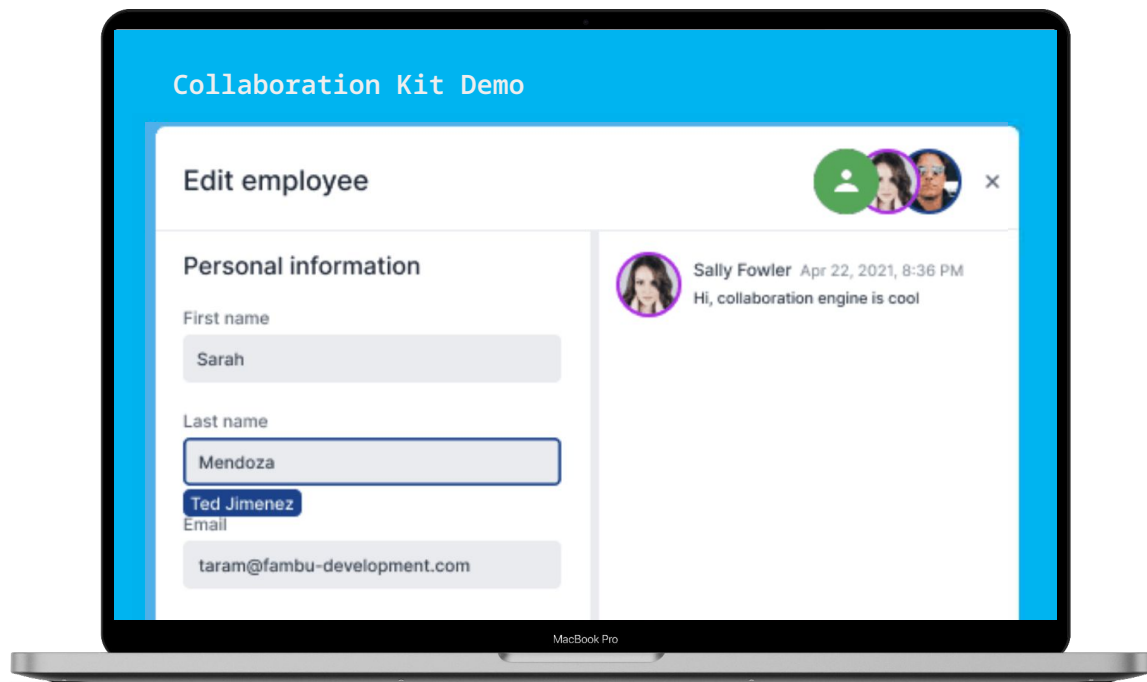
DEMO V

Chat Components



QUICK DETOUR

Vaadin Collaboration Kit



USER PRESENCE

Allow users to see each others presence in a part of the application.

COLLABORATIVE EDITING

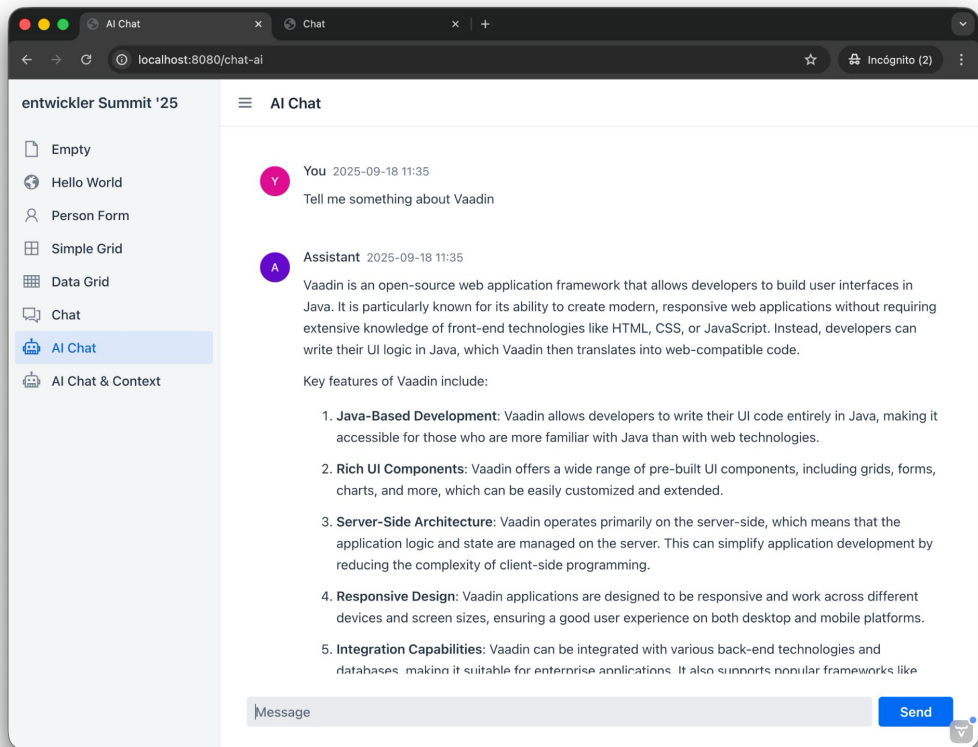
Enable collaborative editing with clear indication of each user's editing location.

INTERACTIVE CHAT

Allow your users to chat with each other in-place for effective collaboration.

DEMO VI

AI Chat Bot



DEMO VII

AI Chat Bot with UI Interaction

The screenshot shows a web application titled "AI Chat with Context" running on a browser at localhost:8080. The application has a sidebar with a menu containing: Empty, Hello World, Person Form, Simple Grid, Data Grid, Chat, AI Chat, and AI Chat & Context (which is highlighted). The main content area is titled "AI Chat with Context" and displays a table of "Conference Talks".

	Category	Topic	Speaker	Language
☐	Backend & Cloud	Serverless Patterns a...	Michael Berger	Deutsch
☒	Backend & Cloud	Kubernetes für Fortge...	Arjun Patel	Englisch
☒	Backend & Cloud	Event-Driven Architec...	Monika Nowak	Englisch
☐	Backend & Cloud	Spring Boot 4: Die ne...	Sebastian Krause	Deutsch
☒	Backend & Cloud	Rust im Backend: Hig...	Olivia Chen	Englisch
☐	Backend & Cloud	Cloud-Native Observa...	Felix Brandt	Deutsch
☒	Backend & Cloud	CQRS & Event Sourci...	Johan Svensson	Englisch

Below the table, a chat interface shows a conversation:

You 2025-09-18 11:36
Highlight all conference talks about Cloud which are in English

Assistant 2025-09-18 11:36
All conference talks about Cloud that are in English have been successfully highlighted. Here are the details:

1. Kubernetes für Fortgeschrittene: Operatoren entwickeln

At the bottom, there is a "Message" input field, a "Send" button, and a "Clear" button.

RETHINKING UI

X – UI Modality for AI Agents

Human Computer Interaction
Computer Human Interaction

Visit the prototype playground at
computerhumaninteraction.com



Where to go from here?

Get started with Vaadin

Create your own starter project
start.vaadin.com

start.spring.io

(add Vaadin dependency)

More starters and templates
(Spring, CDI, Gradle, Quarkus, ...)
github.com/vaadin

Tutorials & Documentations
vaadin.com/start
vaadin.com/docs

Get help
vaadin.com/forum
Github, Stack Overflow

vaadin}>

Where to go from here?

Get started with Vaadin



Conference Slides
Demo Project

[github.com/jtreml/
entwickler-summit-
showcase](https://github.com/jtreml/entwickler-summit-showcase)

Meet us at
Vaadin Create 2025
in Frankfurt
Oct 21-22, 2025

[vaadin.com/vaadin-
create-2025](https://vaadin.com/vaadin-create-2025)

Thank you

**Feel free to
reach out!**

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